

ReactJS Hands-on - Counter App (React State)

Using the React state object with a class component

1. Objective

Create a React app **counterapp** with a component **CountPeople** that tracks how many people entered and exited a mall, using the constructor and **state**. Two methods, `UpdateEntry()` and `UpdateExit()`, update the entry and exit counts. Two buttons, **Login** and **Exit**, trigger these methods.

2. Prerequisites

- Node.js
- NPM
- Visual Studio Code

3. Create the React App

● ● ● Command Prompt

```
C:\Users\student> npx create-react-app counterapp
Creating a new React app in C:\Users\student\counterapp.
Installing react, react-dom, and react-scripts...

Success! Created counterapp at C:\Users\student\counterapp
Happy hacking!

C:\Users\student> cd counterapp
C:\Users\student\counterapp> code .
```

4. CountPeople Component

Added a new file `CountPeople.js` in the **src** folder. The constructor initialises state with `entrycount` and `exitcount` set to 0. `UpdateEntry()` increments the entry count, `UpdateExit()` increments the exit count, and each is wired to its own button.

src/CountPeople.js

```
import React, { Component } from 'react';

class CountPeople extends Component {
  constructor(props) {
    super(props);
    this.state = {
      entrycount: 0,
      exitcount: 0
    };
    this.UpdateEntry = this.UpdateEntry.bind(this);
    this.UpdateExit = this.UpdateExit.bind(this);
  }

  UpdateEntry() {
    this.setState({ entrycount: this.state.entrycount + 1 });
  }

  UpdateExit() {
    this.setState({ exitcount: this.state.exitcount + 1 });
  }

  render() {
    return (
      <div>
        <h2>Mall Entry / Exit Counter</h2>
        <button onClick={this.UpdateEntry}>Login</button>
        <button onClick={this.UpdateExit}>Exit</button>
        <p>Number of people entered: {this.state.entrycount}</p>
        <p>Number of people exited: {this.state.exitcount}</p>
      </div>
    );
  }
}

export default CountPeople;
```

5. App.js

Rendered the `CountPeople` component from `App.js`.

src/App.js

```
import React from 'react';
import CountPeople from './CountPeople';
```

```
function App() {  
  return (  
    <div className="App">  
      <CountPeople />  
    </div>  
  );  
}  
  
export default App;
```

6. Run the Application

Command Prompt

```
C:\Users\student\counterapp> npm start  
Compiled successfully!  
  
You can now view counterapp in the browser.  
Local: http://localhost:3000
```

7. Output

Initial load – both counts start at 0.

localhost:3000

Mall Entry / Exit Counter

Login Exit

Number of people entered: 0

Number of people exited: 0

After clicking **Login** 3 times and **Exit** 1 time – each button updates only its own counter in state, and the component re-renders automatically.

localhost:3000

Mall Entry / Exit Counter

Login Exit

Number of people entered: 3

Number of people exited: 1

8. Observation

Storing entrycount and exitcount in a single state object, initialised in the constructor, lets each button call `setState()` independently. React re-renders only the parts of the UI that depend on the changed state, so clicking Login updates the entry count while the exit count stays untouched, and vice versa.